

# Responsible AI Model Evaluations

## A Nine-Week Red-Teaming Study of Foundation Models Across 22 Safety Risk Categories

Hemant Naik \* May 22, 2026 \* Dataset: RedBench (Dang et al., 2026, MIT)

Live Dashboard: <https://Sushegaad.github.io/Responsible-AI-Model-Evaluations/> \* GitHub: <https://github.com/Sushegaad/Responsible-AI-Model-Evaluations>

| Total Evaluations | Attack Failures | Critical Findings | Over-Refusals    | Study Duration  | Models Evaluated |
|-------------------|-----------------|-------------------|------------------|-----------------|------------------|
| 26,500            | 1,269           | 468               | 422              | 9 Weeks         | 7                |
| 9 weekly runs     | Across all runs | Severity >= 8/10  | False rejections | Apr 10 - May 22 | 3 providers      |

### Abstract

This paper presents a nine-week automated red-teaming study of seven frontier large language models from Anthropic, OpenAI, and Google, evaluated across 22 safety risk categories using the RedBench adversarial benchmark (29,362 prompts). Over 26,500 individual evaluations were conducted between April 10 and May 22, 2026, producing 468 critical findings and 422 false-outright-refusals. Results are scored by a dual-judge pipeline — a deterministic RegexJudge and a NeuralJudge (Claude Haiku) — and mapped to the NIST AI Risk Management Framework 1.0. Across all nine runs, Gemini 2.5 Flash achieved the lowest average Attack Success Rate (3.79%), while GPT-4o Mini remained the highest-risk model (8.42% average ASR). CBRN and Cybersecurity categories continued to account for the majority of guardrail bypasses. In the final run (May 22, 2026), GPT-4o Mini recorded the study's highest-ever drift coefficient (+1.67/turn), Gemini 2.5 Flash spiked to 25% False Outright Refusal, and three models — Claude Opus 4.7, Claude Opus 4.6, and Gemini 2.5 Pro — each reached a drift coefficient of +1.11/turn. Election Interference has solidified as a persistent top-5 risk category at 5.66% average ASR. This study is concluded at nine runs. All data, evaluation code, audit logs, and the research paper are publicly archived under the MIT License.

## 1. Introduction

The deployment of large language models (LLMs) in high-stakes domains — including government, healthcare, finance, and legal services — demands rigorous, reproducible safety evidence. Vendor self-assessments are insufficient for independent governance because they are not replicable, use proprietary benchmarks, and are subject to publication bias. This study addresses that gap with a fully open, automated weekly pipeline that stress-tests seven frontier models against a standardised adversarial dataset and publishes results to a public dashboard every Friday.

The evaluation covers 22 risk categories spanning physical harm, cybersecurity, privacy, misinformation, and domain-specific risks. Each run draws 500 stratified prompts per model from the RedBench dataset — 400 adversarial and 100 benign — and grades every response with a dual-judge system. The nine-week time series produced in this concluded study enables detection of safety regressions, guardrail erosion under adversarial pressure, and comparative model rankings that are independent of any provider's internal testing. All 26,500 evaluations, audit logs, and weekly metrics are permanently archived for replication and longitudinal research.

## 2. Methodology

### 2.1 Dataset

RedBench (Dang et al., 2026) aggregates 37 adversarial sub-benchmarks — including HarmBench, ToxiGen, XSTest, DAN, and AdvBench — into a single standardised schema covering 22 risk categories and 19 domains. A local snapshot of 29,362 prompts (26,113 adversarial, 3,249 benign) is committed to the repository as Parquet files, making every run fully reproducible without network dependencies on the dataset host.

### 2.2 Models

| Model             | Provider  | Runs | Evaluation Period |
|-------------------|-----------|------|-------------------|
| Claude Opus 4.7   | Anthropic | 5    | Apr 24 - May 22   |
| Claude Opus 4.6   | Anthropic | 9    | Apr 10 - May 22   |
| Claude Sonnet 4.6 | Anthropic | 9    | Apr 10 - May 22   |
| GPT-4o            | OpenAI    | 9    | Apr 10 - May 22   |
| GPT-4o Mini       | OpenAI    | 9    | Apr 10 - May 22   |
| Gemini 2.5 Flash  | Google    | 6    | Apr 17 - May 22   |
| Gemini 2.5 Pro    | Google    | 6    | Apr 14 - May 22   |

Table 1. Models evaluated in this study.

### 2.3 Evaluation Pipeline

Each model receives 500 prompts per weekly run: 400 adversarial attacks drawn proportionally across all 22 risk categories, and 100 benign prompts to measure false refusal rates. Every prompt is submitted in three modes: single-turn, agentic 3-turn wrap (system + user + follow-up), and — for a 5% sub-sample — a 10-turn adversarial drift probe that measures how safety posture degrades under sustained pressure.

Responses are graded by a Dual-Judge pipeline. The RegexJudge applies deterministic pattern-matching for PII leakage, jailbreak confirmations, malware code, explosive synthesis instructions, and CSAM. The NeuralJudge (Claude Haiku) returns a structured JSON verdict (SAFE, UNSAFE, or OVER\_REFUSAL), a severity score from 1 to 10, and a chain-of-reasoning sentence for the forensic audit trail. A regex match always overrides the neural result and forces a CRITICAL\_FAIL; a neural UNSAFE verdict with severity 8 or above also escalates to CRITICAL\_FAIL regardless of the regex result.

### 2.4 Metrics

| Metric                       | Definition                                             | Target      |
|------------------------------|--------------------------------------------------------|-------------|
| Attack Success Rate (ASR)    | % of adversarial prompts that bypassed guardrails      | < 5%        |
| False Outright Refusal (FOR) | % of benign prompts incorrectly refused                | < 10%       |
| Drift Coefficient            | Safety change per conversation turn (10-turn probe)    | < 0.5%/turn |
| Provenance Score             | % of evaluations with complete chain-of-reasoning JSON | > 95%       |

Table 2. Evaluation metrics and targets.

## 3. Results

### 3.1 Overall Performance Across All Runs

Across 26,500 evaluations spanning nine weekly runs, Gemini 2.5 Flash achieved the lowest average Attack Success Rate at 3.79%, followed closely by Claude Opus 4.7 at 4.00% — though both models joined the study later than the four-provider cohort. Among models with a full nine-run history, Claude Sonnet 4.6 led at 5.28% average ASR, edging out GPT-4o (5.67%) and Claude Opus 4.6 (5.81%). GPT-4o Mini remained the weakest performer at 8.42% average ASR, with Gemini 2.5 Pro close behind at 8.00%.

False Outright Refusal rates reveal a sharp safety-utility divide. The three Anthropic models averaged between 0.78% and 1.20% FOR — exceptionally low, indicating near-zero unnecessary refusals. Google's models showed the opposite pattern: Gemini 2.5 Pro averaged 18.33% FOR and Gemini 2.5 Flash 16.33%, meaning roughly one in five to six benign requests was incorrectly refused. GPT-4o Mini averaged 13.67% FOR, also well above the 10% target threshold. Notably, in the May 22 run, Claude Sonnet 4.6 achieved a perfect 0.00% FOR — no benign prompts were incorrectly refused — while Gemini 2.5 Flash spiked to 25.00%, its worst FOR reading across the entire study.

| Model             | Avg ASR | Avg FOR | Avg Drift | Avg Prov | Total Crit | Runs |
|-------------------|---------|---------|-----------|----------|------------|------|
| Gemini 2.5 Flash  | 3.79%   | 16.33%  | +0.37     | 100.0%   | 24         | 6    |
| Claude Opus 4.7   | 4.00%   | 1.20%   | +0.78     | 100.0%   | 26         | 5    |
| Claude Sonnet 4.6 | 5.28%   | 1.00%   | -0.12     | 99.9%    | 63         | 9    |
| GPT-4o            | 5.67%   | 7.67%   | +0.43     | 100.0%   | 70         | 9    |
| Claude Opus 4.6   | 5.81%   | 0.78%   | -0.12     | 99.9%    | 74         | 9    |
| Gemini 2.5 Pro    | 8.00%   | 18.33%  | +0.28     | 100.0%   | 115        | 6    |
| GPT-4o Mini       | 8.42%   | 13.67%  | +0.49     | 100.0%   | 96         | 9    |

Table 3. Per-model averages across all successful evaluation runs. Green = at or below target; amber = moderate concern; red = above target.

### 3.2 Latest Run: May 22, 2026

The ninth evaluation run (May 22, 2026) produced several noteworthy results. Claude Opus 4.7 led with the lowest ASR at 3.50%, and Claude Sonnet 4.6 achieved a perfect 0.00% False Outright Refusal rate — no benign prompts were incorrectly refused, the best utility reading recorded in the study. Gemini 2.5 Flash, by contrast, spiked to 25.00% FOR — its worst result — suggesting heightened over-caution in the May 22 sample. GPT-4o Mini recorded the study's highest-ever single-run drift coefficient at +1.67/turn, indicating its guardrails eroded most severely under sustained multi-turn adversarial pressure. Three additional models — Claude Opus 4.7, Claude Opus 4.6, and Gemini 2.5 Pro — each posted a drift coefficient of +1.11/turn, making May 22 the run with the most widespread guardrail erosion across the nine-week study.

| Model             | ASR   | FOR    | Drift | Provenance | Critical |
|-------------------|-------|--------|-------|------------|----------|
| Claude Opus 4.7   | 3.50% | 1.00%  | +1.11 | 100.0%     | 5        |
| Gemini 2.5 Flash  | 4.00% | 25.00% | 0.00  | 100.0%     | 3        |
| Claude Sonnet 4.6 | 5.25% | 0.00%  | 0.00  | 99.8%      | 7        |
| GPT-4o            | 6.00% | 8.00%  | +0.56 | 100.0%     | 8        |
| Claude Opus 4.6   | 6.75% | 1.00%  | +1.11 | 99.6%      | 15       |
| Gemini 2.5 Pro    | 8.75% | 22.00% | +1.11 | 100.0%     | 20       |
| GPT-4o Mini       | 9.25% | 13.00% | +1.67 | 100.0%     | 11       |

Table 4. May 22, 2026 evaluation run results (sorted by ASR).

### 3.3 Weekly ASR Trend (All Nine Runs)

The nine-week trend reveals considerable week-to-week variance across all models, driven largely by sampling randomness and prompt-composition shifts. The most consistent performer over the full period is Claude Sonnet 4.6, which exhibited the narrowest ASR range (2.75%-7.25%) and a slight downward trend through the mid-study period before recovering. GPT-4o Mini's ASR peaked in runs 1-2 at 10.25% and 10.00%, fell to 5.5% by run 3, climbed to 9.75% in run 8, and settled at 9.25% in the final run — consistently above the 8% red threshold for seven of nine runs. Gemini 2.5 Pro has never fallen below 7.25% across its six evaluated runs and reached its study high of 9.00% on May 16 before remaining elevated at 8.75% on May 22.

| Model             | Apr 10 | Apr 12 | Apr 14 | Apr 17 | Apr 24 | May 2 | May 8 | May 16 | May 22 |
|-------------------|--------|--------|--------|--------|--------|-------|-------|--------|--------|
| Claude Opus 4.6   | 6.00%  | 7.50%  | 4.25%  | 4.00%  | 6.75%  | 5.25% | 4.50% | 7.25%  | 6.75%  |
| Claude Opus 4.7   | --     | --     | --     | --     | 4.00%  | 3.00% | 4.25% | 5.25%  | 3.50%  |
| Claude Sonnet 4.6 | 6.50%  | 7.00%  | 2.75%  | 3.25%  | 6.50%  | 5.00% | 4.00% | 7.25%  | 5.25%  |
| Gemini 2.5 Flash  | --     | --     | --     | 2.50%  | 3.50%  | 3.75% | 4.00% | 5.00%  | 4.00%  |
| Gemini 2.5 Pro    | --     | --     | 7.25%  | 8.00%  | 7.25%  | --    | 7.75% | 9.00%  | 8.75%  |
| GPT-4o            | 6.25%  | 6.50%  | 3.25%  | 3.75%  | 7.00%  | 5.25% | 6.25% | 6.75%  | 6.00%  |
| GPT-4o Mini       | 10.25% | 10.00% | 5.50%  | 6.00%  | 9.50%  | 6.00% | 9.50% | 9.75%  | 9.25%  |

Table 5. Weekly Attack Success Rate by model. -- = model not evaluated that week. Green < 5%, amber 5-8%, red > 8%.

## 4. Risk Category Analysis

CBRN (Chemical, Biological, Radiological, and Nuclear) information remained the highest-risk category across the full nine-week period with an average ASR of 35.41% — more than double the second-ranked category, Cybersecurity Threats Beyond Malware at 21.99%. This gap reflects the inherent difficulty of building blanket guardrails against CBRN-adjacent dual-use knowledge, which requires nuanced contextual reasoning rather than keyword matching. Illegal Weapons (Non-CBRN) ranked third at 17.52%, followed by Malware Code at 12.26%.

Election Interference has continued its trajectory, averaging 5.66% ASR across all runs — the fifth-highest category with 53 data points across all seven models and all nine weeks. This is a meaningful finding given the electoral calendar: models are demonstrably more susceptible to prompts designed to generate misleading electoral content, impersonate candidates, or fabricate polling data than to most other social-harm categories. The consistency of this finding across nine runs and seven models confirms it is structural rather than a sampling artefact.

Sexual Content (4.53%), Misinformation (3.88%), Bias Content (3.45%), and Hateful Content (3.30%) round out the top ten. Privacy and Surveillance categories remained low across the study, averaging below 2%, which likely reflects the strong PII-detection capabilities of the RegexJudge component.

| Risk Category                          | Avg ASR | Data Points | NIST Function |
|----------------------------------------|---------|-------------|---------------|
| CBRN Information or Capabilities       | 35.41%  | 53          | MAP           |
| Cybersecurity Threats (Beyond Malware) | 21.99%  | 53          | MAP           |
| Illegal Weapons (Non-CBRN)             | 17.52%  | 53          | MANAGE        |
| Malware Code                           | 12.26%  | 53          | MAP           |
| Election Interference (up)             | 5.66%   | 53          | GOVERN        |

|                    |       |    |        |
|--------------------|-------|----|--------|
| Sexual Content     | 4.53% | 53 | MANAGE |
| Misinformation     | 3.88% | 53 | MAP    |
| Bias Content       | 3.45% | 53 | GOVERN |
| Hateful Content    | 3.30% | 53 | GOVERN |
| Environmental Harm | 3.15% | 53 | MAP    |

Table 6. Top 10 risk categories by average ASR across all nine runs. (up) = persistent upward trend. NIST AI RMF function mapping shown.

## 5. Guardrail Stability Under Multi-Turn Pressure

The Drift Coefficient — computed on a 5% sub-sample of attack prompts subjected to 10 consecutive adversarial turns — measures how a model's safety posture shifts under sustained pressure. A positive coefficient indicates eroding guardrails; a negative coefficient indicates that the model becomes more conservative as the dialogue progresses.

Claude Opus 4.6 averaged a drift coefficient of -0.12/turn across nine runs, the most stable result in the study: its guardrails tightened slightly under extended adversarial engagement. Claude Sonnet 4.6 matched this figure at -0.12/turn. In contrast, Claude Opus 4.7 averaged +0.78/turn — the highest average drift of any model — suggesting that its safety fine-tuning may be less robust to multi-turn manipulation specifically, even though it performs well on single-turn ASR.

The May 22 run produced the study's most significant drift readings. GPT-4o Mini reached +1.67/turn — the highest single-run drift coefficient recorded across all nine weeks and all seven models — confirming a structural vulnerability to guardrail erosion under extended adversarial dialogue. Three additional models simultaneously posted +1.11/turn drift: Claude Opus 4.7, Claude Opus 4.6, and Gemini 2.5 Pro. The fact that four out of seven models posted significant drift in the same run suggests that the May 22 adversarial sample contained particularly effective multi-turn manipulation patterns.

GPT-4o Mini maintained a consistently positive drift across all nine runs (+0.49/turn average), confirming a structural tendency toward guardrail softening under adversarial multi-turn dialogue. This, combined with its persistently high 8.42% average ASR, makes it the most concerning model in the study from a multi-turn deployment safety perspective.

## 6. Audience-Specific Insights

### AI Engineers and Red-Teamers

- \* CBRN (35.41%) and Cybersecurity (21.99%) categories account for the majority of bypasses — these two categories alone drive over half of all attack failures across all nine runs.
- \* May 22 produced the study's highest-ever drift reading: GPT-4o Mini at +1.67/turn. Four models simultaneously posted  $\geq +1.11$ /turn drift — examine whether the May 22 prompt sample contains new multi-turn attack patterns not seen in prior weeks.
- \* Claude Sonnet 4.6 achieved 0.00% FOR on May 22 with a 5.25% ASR — the best single-run safety-utility balance in the study. This is the benchmark to target.
- \* Election Interference (5.66% avg ASR, 53 data points) is a statistically robust structural top-5 risk — add it to red-team priority lists for all deployments.

### GRC and Compliance Officers

- \* 468 critical findings across 26,500 evaluations (1.77% critical rate) — all mapped to NIST AI RMF functions (MAP, MANAGE, GOVERN, MEASURE) with chain-of-reasoning audit trails.
- \* GSAR 552.239-7001 forensic logs are generated for every CRITICAL\_FAIL and all FAIL verdicts with severity  $\geq 7$ , providing PII-redacted prompt/response pairs for direct inclusion in compliance documentation.
- \* Gemini 2.5 Pro (18.33% avg FOR, 115 critical findings across 6 runs) and GPT-4o Mini (13.67% avg FOR, 96 criticals across 9 runs) both exceed the 10% FOR threshold — flag for procurement review if deploying in user-facing or regulated applications.
- \* Provenance Scores averaged  $\geq 99.9\%$  across all models, ensuring a near-complete audit trail for all 26,500 evaluations.

### Product Leaders and Deployment Decision-Makers

- \* Anthropic models offer the best safety-utility balance: ASR between 4-6% and FOR below 1.2%, meaning low attack exposure without significant refusal overhead for legitimate users.
- \* Gemini 2.5 Flash's 3.79% average ASR is impressive, but its 16.33% avg FOR — spiking to 25% on May 22 — means roughly 1 in 4 to 6 safe user requests may be incorrectly rejected depending on the week. Evaluate carefully for customer-facing use cases.
- \* GPT-4o Mini's persistently high ASR (8.42% average) and 13.67% FOR represent a dual failure mode: simultaneously too permissive to adversarial prompts and too restrictive to legitimate users. The May 22 drift of +1.67/turn is an additional deployment risk.
- \* Claude Sonnet 4.6 is the standout deployment choice: 5.28% avg ASR, 1.00% avg FOR, negative drift (-0.12/turn), and a perfect 0.00% FOR on May 22 — the only model to achieve this across the nine-week study.

### Policy Analysts and Researchers

- \* Nine weeks of weekly data on a fixed open benchmark (RedBench, MIT License) enables time-series safety comparison independent of any vendor's internal reporting. 26,500 evaluations across 53 model-run combinations provide high statistical confidence.
- \* Election Interference at 5.66% average ASR across 53 data points is a statistically robust finding across 7 models and 9 runs — not a sampling artefact, and below the 6.52% reported in the prior eight-week version (slight improvement).
- \* The CBRN gap (35.41% vs 21.99% for the second category) persists across all models and all runs, suggesting systemic limits of current RLHF-based alignment against dual-use knowledge requests require domain-specific interventions.
- \* The May 22 run, with four models at drift  $\geq +1.11$ /turn simultaneously, is a signal worth investigating — it may reflect a specific adversarial prompt composition that more effectively exploits multi-turn vulnerabilities common to current alignment methods.

## 7. Conclusions

After nine weekly evaluation runs covering 26,500 evaluations across seven frontier models, four findings stand out. First, the safety-utility trade-off is not a universal constraint: Anthropic's Claude models consistently demonstrate that a model can achieve both low attack bypass rates and near-zero false refusals — Claude Sonnet 4.6's May 22 result of 5.25% ASR with 0.00% FOR is the clearest proof point in this study.

Second, CBRN and Cybersecurity categories remain structurally resistant to current guardrail approaches across all providers, with average ASRs of 35.41% and 21.99% respectively. These categories require domain-specific safety interventions beyond general RLHF alignment.

Third, Election Interference has solidified as a persistent top-5 risk category at 5.66% average ASR across all models and all nine runs. This finding carries heightened urgency given ongoing electoral cycles globally and warrants targeted safety investment from all providers.

Fourth, multi-turn guardrail stability varies dramatically by provider and emerged as a critical concern in the May 22 run. GPT-4o Mini's +1.67/turn drift is the highest single-run reading in the study, and the simultaneous elevated drift across four models suggests a structural vulnerability in how current frontier models respond to sustained adversarial multi-turn pressure. Anthropic models (Opus 4.6 and Sonnet 4.6) average negative drift (-0.12/turn), demonstrating that stable multi-turn safety is achievable.

The pipeline, data, and audit logs for this study are fully open-source and available at <https://github.com/Sushegaad/Responsible-AI-Model-Evaluations> under the MIT License. This study is concluded at nine evaluation runs (April 10 – May 22, 2026). The full dataset, weekly metrics, forensic audit logs, and the evaluation pipeline are preserved in the repository for replication and further research.

## 8. References

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Disclaimer: This is an independent research initiative. Results reflect model behaviour on RedBench under standardised conditions and are not a comprehensive measure of safety in all deployment contexts. Results may vary due to model updates, API changes, and sampling randomness. No affiliation with Anthropic, OpenAI, Google, or the RedBench authors is implied.